

Rajiv V. Joshi

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Education

Carnegie Mellon University	Pittsburgh, PA
Master of Science in Mechanical Engineering - Research GPA: 4.0/4.0	May 2026
Virginia Tech	Blacksburg, VA
Bachelors of Science in Mechanical Engineering - Robotics and Mechatronics GPA: 3.64/4.0	May 2024
Bachelors of Science in Mechanical Engineering - Mechanical GPA: 3.64/4.0	May 2024

Coursework

Masters: ADV Mechatronics (TA) | Haptic Design Interfaces | Modern Control Theory | Computer Vision for Engineers (In-Progress) | ML and AI for Engineers (In-Progress)

Undergraduate: Robotics and Automation | Kinematics | Differential Equations | Deforms | Dynamics

Skills

Embedded Systems: Arduino | STM 32 | Jetson Orin Nano | Oscilloscope | DMM | Serial Communication (SPI, I2C, CAN)

Manufacturing: PCB Design | Soldering | Circuit Schematics | Rapid Prototyping | 3-D printing | Machining

Software/Programming: Python | C/C++ | MATLAB | SolidWorks | Fusion 360 | KiCAD | ANSYS | Linux Command Line

General: Hardware/Software Debugging, Firmware programming, Fast-paced environments, Cross-team communication

Research Experience

MetaMobility Lab at Carnegie Mellon University	Pittsburgh, PA
<i>Knee Exoskeleton – MS Research Project</i>	April 2025 - Present

- Coordinated and led a team of 5 students in research, design, and manufacturing a lightweight knee exoskeleton capable of delivering 20 Nm of torque directly to knee joints for people suffering from osteoarthritis or gait impairment
- Utilized a Teensy 4.1 to implement **SPI** protocol, in **C/C++**, with **IMU sensors** using CAT6 connectors and twisted pair cables to **shield signal from noise**, increasing the gait cycle estimator loop frequency from 100Hz to 500Hz
- Debugged multiple hardware and firmware faults using an **Oscilloscope and DMM** to **visualize logic signals**, voltage values, and **circuit continuity** to rapidly diagnose causes and **iterate/manufacture immediate solutions**

MetaMobility Lab at Carnegie Mellon University	Pittsburgh, PA
<i>Hip Exoskeleton – MS Research</i>	May 2024 - Present

- Optimized hip exoskeleton hardware for improved fitting to users using **human biomechanics** concepts, **SolidWorks**, and **3D printing**, yielding comfortable motor-joint alignment and 25% reduced system weight
- Redesigned circuit schematics and **PCB designs** to integrate embedded systems such as **MCUs**, **IMUs**, and **FSRs** via **KiCAD**, reducing connections from 8 to 4 while maintaining robust connections and signal integrity
- Integrated **CAN bus** and **I2C** protocols, in **Python** and **C/C++**, with motors encoders and **IMUs**, enabling gait cycle estimation for controllers and hardware torque control upwards of 18 Nm, cutting user hip work by 20% to 30%
- Communicated with various **global manufacturers** for pricing and outsourcing manufacturing of Carbon Fiber and Nylon based parts designed in **SolidWorks** considering basic **DFM** and **DFA** concepts

Academic Projects

Carnegie Mellon University	Pittsburgh, PA
<i>Haptic Design Interfaces: 2 DoF Haptic Hand Interface – Course Project</i>	April 2025 - May 2025

- Developed a open-source compact haptic fingers device, with a team of 6 graduate students, meshing **STM32 MCU**, 4 low torque motors, and 3D-printed parts as a teaching tool for **robot kinematics**, **biological sensors**, and **haptics**
- Visualized a finger joint workspace in **MATLAB** to define the mechanical operational range of a capstan-driven end-effector, which was controlled in **C/C++** using **PID** and **forward kinematics** to successfully render virtual 3D objects.

Carnegie Mellon University	Pittsburgh, PA
<i>ADV Mechatronics: Embedded System Game Design – Course Project</i>	November 2024 - December 2024

- Implemented **PWM**, **timer interrupts**, and **event driven state-machine** on an **STM32** for controlling a custom **H-bridge motor driver circuit**, in whack-a-mole style arcade game designed with a team of 3.
- Created detailed circuit schematics, using KiCAD, for guiding protoboard **soldering/manufacturing**, optimizing cable routing between components, and assisted in **debugging circuit/logic** issues with an **oscilloscope** and **DMM**